

PROBLEM 2 (1 hour)

As a consequence of an underwater explosion, a spherical cavity of initial radius R_0 and initial radial velocity V_0 is created inside the ocean. The growth of the cavity induces a flow in the surrounding liquid, whose density ρ and viscosity μ can be considered constant. It is also assumed that the effect of gravity is negligible, and that the motion is spherically symmetric, $\bar{v} = v_r(r, t) \bar{e}_r$, where $R_c(t) \leq r < \infty$, being $R_c(t)$ the radius of the cavity at time t . The liquid is at rest far from the cavity, $r \rightarrow \infty$, where the pressure is p_∞ , while the pressure inside the cavity can be neglected compared to p_∞ .

1. (1.5 points). Find the radial velocity of the liquid, $v_r(r, t)$, as a function of the radius of the cavity, $R_c(t)$, and of its associated expansion velocity, $\dot{R}_c(t) = dR_c/dt$. Show that the resulting flow is irrotational.
2. (1.5 points). Obtain the pressure field, $p(r, t)$, as a function of $R_c(t)$, $\dot{R}_c(t)$, $\ddot{R}_c(t)$, ρ and p_∞ .
3. (1 point). Evaluate the resulting pressure distribution at $r = R_c(t)$, to find the ordinary differential equation (ODE) with initial conditions satisfied by $R_c(t)$, in terms of the unknown liquid pressure at the cavity boundary, $p_R(t)$.
4. (1.5 points). Find $p_R(t)$ by imposing the condition of null stress of the liquid at the cavity wall, $\bar{\tau} \cdot \bar{n}|_{r=R_c(t)} = 0$, where $\bar{\tau} = -p\bar{I} + \bar{\tau}'$ is the stress tensor.
5. (1.5 points). Using the answer to question #4, write the ODE obtained in question #3 in closed form. Nondimensionalize the corresponding initial value problem, using the following dimensionless variables: $R = R_c/R_0$, $\tau = t V_0/R_0$. Show that the problem is governed by two dimensionless parameters, namely the Reynolds number, $Re = \rho V_0 R_0/\mu$, and $\Lambda = p_\infty/(\rho V_0^2)$.
6. (1.5 points). Simplify the ODE in the limit $Re \rightarrow \infty$. By using $z = \ln(R)$ as the independent variable and $F = (dR/d\tau)^2$ as the dependent one, show that $F(z)$ satisfies the linear ODE

$$\frac{dF}{dz} + 3F = -2\Lambda,$$

with initial condition $F(0) = 1$. Obtain $F(z)$, and use it to find $dR/d\tau$ as a function of R .

7. (1.5 points). Obtain the maximum radius of the cavity, $R_{\max}$, as a function of Λ , and show that the time $\tau_{\max}$, defined such that $R(\tau_{\max}) = R_{\max}$ is given by the integral

$$\tau_{\max}(\Lambda) = \int_1^{\left(1+\frac{3}{2\Lambda}\right)^{1/3}} \frac{dx}{\sqrt{x^{-3} + \frac{2\Lambda}{3}(x^{-3} - 1)}}.$$

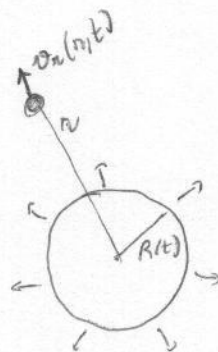
Provide an explicit approximation for $\tau_{\max}$ in the limit $\Lambda \ll 1$.

8. (1 additional point). Simplify the ODE found in question #5 in the limit $Re \rightarrow 0$. Is it possible to satisfy the initial conditions? Find an approximate solution, and discuss its validity.

$$(1) \nabla \cdot \vec{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) = 0 \Rightarrow \boxed{r^2 v_r = A(t)}$$

AT $r = R_c(t)$: $R_c^2 v_r(R) = A(t)$, AND $v_r(R) = \frac{dR_c}{dt} = \dot{R}_c$

$$\Rightarrow \boxed{A(t) = R_c^2 \dot{R}_c} \Rightarrow \boxed{v_r(r, t) = \frac{R_c^2 \dot{R}_c}{r^2}} \rightarrow \boxed{\nabla \wedge \vec{v} = 0} \text{ (IRROTATIONAL)}$$



$$(2) \frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} = -\nabla p - \mu \nabla \wedge (\nabla \wedge \vec{v})$$

SINCE $\nabla \wedge \vec{v} = 0$ (IRROTATIONAL FLOW), $\frac{\partial \vec{v}}{\partial t} + \nabla \left(\frac{p}{\rho} + \frac{1}{2} |\vec{v}|^2 \right) = 0 \Rightarrow$

$$\Rightarrow \frac{\partial v_r}{\partial t} + \frac{2}{r} \left(\frac{p}{\rho} + \frac{v_r^2}{2} \right) = 0 \Rightarrow \frac{p}{\rho} + \frac{v_r^2}{2} = - \int \frac{\partial v_r}{\partial t} dr + B(t)$$

$$\left. \begin{aligned} \frac{\partial v_r}{\partial t} &= \frac{1}{r^2} (R_c^2 \ddot{R}_c + 2 R_c \dot{R}_c^2) \\ \frac{v_r^2}{2} &= \frac{1}{2} \frac{R_c^4 \dot{R}_c^2}{r^4} \end{aligned} \right\} \begin{aligned} \frac{p}{\rho} &= - \frac{1}{2} \frac{R_c^4 \dot{R}_c^2}{r^4} + \frac{R_c^2 \ddot{R}_c + 2 R_c \dot{R}_c^2}{r} + B(t) \\ \xrightarrow{r \rightarrow \infty} \boxed{\frac{p_\infty}{\rho} = B(t)} &\Rightarrow \boxed{\frac{p - p_\infty}{\rho} = \frac{R_c^2 \ddot{R}_c + 2 R_c \dot{R}_c^2}{r} - \frac{R_c^4 \dot{R}_c^2}{2 r^4}} \end{aligned}$$

(3) AT $r = R(t)$, WHERE $p = p_R(t)$: $\boxed{R_c \ddot{R}_c + \frac{3}{2} \dot{R}_c^2 = \frac{p_R(t) - p_\infty}{\rho}}$
WITH INITIAL CONDITIONS $\rightarrow \boxed{R_c(0) = R_0, \dot{R}_c(0) = V_0}$

(4) $\vec{c} \cdot \vec{n} = -p \vec{n} + \vec{c}' \cdot \vec{n} = 0$ AT $r = R_c(t)$ (CAVITY GAS AT $p_g \approx 0$ AND WITH $\vec{c}'_g \approx 0$)
 $\vec{c}' \cdot \vec{n} = \vec{c}' \cdot \vec{e}_r = \tau'_{rr} \vec{e}_r + \tau'_{r\theta} \vec{e}_\theta + \tau'_{r\phi} \vec{e}_\phi$ ($\tau'_{r\theta} = \tau'_{r\phi} = 0$)
AND $\tau'_{rr} = 2\mu \frac{\partial v_r}{\partial r} = - \frac{4\mu R_c^2 \dot{R}_c}{r^3} \Rightarrow \vec{c}' \cdot \vec{n} \Big|_{r=R} = - \frac{4\mu R_c^2 \dot{R}_c}{R_c^3} \vec{e}_r = -4\mu \frac{\dot{R}_c}{R_c} \vec{e}_r \Rightarrow$
 $\Rightarrow \vec{c} \cdot \vec{n} \Big|_{r=R(t)} = - (p_R(t) + 4\mu \frac{\dot{R}_c}{R_c}) \vec{e}_r = 0 \Rightarrow \boxed{p_R(t) = -4\mu \frac{\dot{R}_c}{R_c}}$

(5) $R_c \ddot{R}_c + \frac{3}{2} \dot{R}_c^2 + 4\mu \frac{\dot{R}_c}{R_c} = - \frac{p_\infty}{\rho}$ WHERE $\boxed{\Lambda = \frac{\mu}{\rho}}$. INTRODUCING $R = \frac{R_c}{R_0}$, $\tau = \frac{V_0}{R_0} t$,
 $R_c(0) = R_0$
 $\dot{R}_c(0) = V_0$
THE EQUATION FOR $R(\tau)$ WRITES:

$$\boxed{RR'' + \frac{3}{2} R'^2 + \frac{4}{Re} \frac{R'}{R} = -\Lambda}$$

WHERE $R' \equiv \frac{dR}{d\tau}$; $Re = \frac{V_0 R_0}{\mu}$ AND $\Lambda = \frac{p_\infty}{\rho V_0^2}$
ARE THE ONLY PARAMETERS

⑥ DEFINING $\begin{cases} z = \ln(R) \\ F(z) = \left(\frac{dR}{dz}\right)^2 \end{cases}$ AND LETTING $Re \rightarrow \infty$:

$$\begin{cases} R \frac{d^2 R}{dz^2} + \frac{3}{2} \left(\frac{dR}{dz}\right)^2 = -\Lambda \\ R(0) = 1 \\ \frac{dR}{dz}(0) = 1 \end{cases} \quad \begin{aligned} &\frac{dF}{dz} = \frac{dF}{dR} \frac{dR}{dz} = 2R \frac{d^2 R}{dz^2} \\ &\hookrightarrow \frac{dF}{dz} \frac{dz}{dR} = 2 \frac{d^2 R}{dz^2} \end{aligned}$$

$z=0: R=1 \Rightarrow F=0$
 $\frac{dR}{dz}=1 \Rightarrow F=1$

$$\begin{cases} \frac{dF}{dz} + 3F = -2\Lambda \\ F(0) = 1 \end{cases}$$

$$F = e^{-3z} + \frac{2\Lambda}{3} (e^{-3z} - 1)$$

$$\downarrow$$

$$\left(\frac{dR}{dz}\right)^2 = R^{-3} + \frac{2\Lambda}{3} (R^{-3} - 1) \Rightarrow \left[\frac{dR}{dz} = \left[R^{-3} + \frac{2\Lambda}{3} (R^{-3} - 1)\right]^{1/2}\right]$$

⑦ MAXIMUM RADIUS R_{max} ACHIEVED WHEN $\left.\frac{dR}{dz}\right|_{z_{max}} = 0 \Rightarrow$

$$\Rightarrow 1 + \frac{2\Lambda}{3} (1 - R_{max}^3) = 0 \Rightarrow R_{max} = \left(1 + \frac{2\Lambda}{3}\right)^{1/3}$$

WHILE z_{max} IS GIVEN BY

$$\frac{dR}{\sqrt{R^{-3} + \frac{2\Lambda}{3} (R^{-3} - 1)}} = dz \quad \begin{aligned} &\int_1^{R_{max}} \frac{dR}{\sqrt{R^{-3} + \frac{2\Lambda}{3} (R^{-3} - 1)}} = z_{max} \\ &\int_1^{R_{max}} \frac{dR}{\sqrt{R^{-3} + \frac{2\Lambda}{3} (R^{-3} - 1)}} = \int_1^{R_{max}} \frac{dR}{\sqrt{R^{-3} + \frac{2\Lambda}{3} (R^{-3} - 1)}} \end{aligned}$$

$$z_{max}(\Lambda) = \int_1^{(1+\frac{2\Lambda}{3})^{1/3}} \frac{dx}{\sqrt{x^{-3} + \frac{2\Lambda}{3} (x^{-3} - 1)}}$$

WHEN $\Lambda \ll 1$ ONE MAY APPROXIMATE

$$z_{max} \approx \int_1^{(\frac{3}{2\Lambda})^{1/3}} x^{3/2} dx = \frac{2}{5} \left[\left(\frac{3}{2\Lambda}\right)^{5/6} - 1 \right]$$

$$\Rightarrow z_{max} \approx \frac{2}{5} \left(\frac{3}{2}\right)^{5/6} \Lambda^{-5/6}$$

⑧ $Re \rightarrow 0$: $\frac{dR}{dz} \approx -\frac{\Lambda Re}{4} R \rightarrow$ CANNOT SATISFY BOTH INITIAL CONDITIONS $R(0) = R'(0) = 1$

(INITIAL TRANSIENT WHERE INERTIA IS ALWAYS IMPORTANT)

$$R(z) \approx e^{-\frac{\Lambda Re}{4} z}$$

→ DOESN'T ACCOMPLISH INITIAL VELOCITY, SO THIS ISN'T VALID DURING THE INITIAL TRANSIENT WHERE INERTIAL TERMS MATTER